

DAX MEASURES DOCUMENTATION

Project Title:

Retail & E-Commerce Sales Performance Analysis

1. Objective:

The objective of this document is to explain the DAX (Data Analysis Expressions) measures created in Power BI for calculating key business metrics. These measures provide dynamic calculations that update automatically based on slicers and filters applied in the dashboard.

2. Tool Used:

- Microsoft Power BI
- DAX (Data Analysis Expressions)

3. Introduction:

DAX (Data Analysis Expressions) is a formula language used in Microsoft Power BI to create custom calculations and measures. DAX measures help summarize data, calculate KPIs, and enable interactive reporting. The measures created for this project allow users to analyze sales performance, profitability, order volume, product returns, and other important business metrics.

4. DAX Measures Created;

Measure 1

Measure Name:

Total Sales

DAX Formula:

Total Sales = SUM(Orders[Sales])

Purpose:

Calculates the total revenue generated from all sales transactions.

Output:

Displays the overall sales value in the KPI card and dashboard visuals.

Measure 2

Measure Name:

Total Profit

DAX Formula:

Total Profit = SUM(Orders[Profit])

Purpose:

Calculates the total profit earned from all completed sales.

Output:

Displays total profit generated by the business.

Measure 3**Measure Name:**

Total Orders

DAX Formula:

Total Orders = DISTINCTCOUNT(Orders[Order ID])

Purpose:

Calculates the total number of unique customer orders.

Output:

Displays the total order count.

Measure 4**Measure Name:**

Total Quantity Sold

DAX Formula:

Total Quantity Sold = SUM(Orders[Quantity])

Purpose:

Calculates the total number of products sold.

Output:

Displays the total quantity sold.

Measure 5**Measure Name:**

Returned Orders

DAX Formula:

Returned Orders = COUNTROWS>Returns)

Purpose:

Calculates the total number of returned orders.

Output:

Displays the total returned orders.

Measure 6**Measure Name:**

Return Rate

DAX Formula:

Return Rate =

`DIVIDE([Returned Orders],[Total Orders],0)`

Purpose:

Calculates the percentage of returned orders compared to total orders.

Output:

Displays the return percentage as a KPI.

5. Purpose of DAX Measures

The DAX measures were created to provide dynamic calculations throughout the dashboard. These measures automatically respond to user selections made through slicers such as Category, Region, Customer Segment, and Payment Method.

The calculated measures improve dashboard interactivity and enable users to analyze business performance from different perspectives without modifying the underlying dataset.

6. Dashboard KPIs Generated Using DAX

The following KPIs were created using DAX measures:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity Sold
- Returned Orders
- Return Rate

These KPIs provide a quick overview of the overall business performance and support decision-making.

7. Advantages of Using DAX Measures

- Dynamic calculations based on filters.
- Improved dashboard performance.
- Real-time KPI updates.
- Better business analysis.

- Reusable calculations across multiple visuals.
- Simplifies complex calculations.

8. Conclusion

DAX measures played an important role in developing the Retail & E-Commerce Sales Performance Analysis dashboard. They enabled accurate calculation of key business metrics and enhanced dashboard interactivity. The measures automatically respond to user selections, making the dashboard more flexible, informative, and suitable for business decision-making.